

PREEKSHIT SAKLANI

B.Tech CSE (AI) | Sushant University, Gurugram

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Gurugram, Haryana, India | Roll No.: 220BTCCSE035

PROFESSIONAL SUMMARY

Final-year B.Tech student specializing in Artificial Intelligence at Sushant University, Gurugram, with demonstrated expertise in designing and deploying end-to-end AI/ML systems, autonomous multi-agent pipelines, and full-stack intelligence platforms. Experienced in building production-grade applications that integrate Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, on-device AI inference, and advanced deep learning architectures. Proven track record of engineering systems from ideation to deployment across Python, Kotlin, Java, TypeScript, and SQL ecosystems. Recipient of multiple hackathon awards, a published researcher, and a Mumbai Hacks HealthTech national finalist. Actively building towards AI/ML Engineer, Deep Learning Engineer, and MLOps Engineer roles, with a strong philosophical commitment to making intelligence accessible without expensive infrastructure.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, SQL, TypeScript, JavaScript, HTML/CSS

AI / ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, LangChain, LangGraph, NLTK, spaCy, OpenCV, LightGBM, XGBoost

LLMs & Agents: GPT-2 (fine-tuning), LLaMA (llama.cpp, GGUF Q4_K_M), Whisper (whisper.cpp), Gemini Pro 1.5, NVIDIA NIM, Ollama, Multi-Agent Systems

RAG & Vector Databases: Pinecone, TF-IDF, In-Memory Vector Search, Embedding Pipelines, Semantic Search

Databases: PostgreSQL, MongoDB, SQLite, Neo4j (Knowledge Graphs), Pinecone

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, SQLAlchemy

Frontend: React, Next.js, TypeScript, Tailwind CSS

DevOps & Cloud: Docker, Kubernetes, Git/GitHub, CI/CD

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Tableau, TensorBoard

Deep Learning Concepts: CNNs, RNNs, Transformers, GANs (StyleGAN), Vision Transformers (ViT), Autoencoders, Transfer Learning, RLDA, SVD

NLP & Text AI: Sentiment Analysis, TF-IDF, Named Entity Recognition, Text Classification, Language Modeling, Tokenization, Lemmatization

Soft Skills: Systems Thinking, Problem Decomposition, Technical Writing, Public Speaking, Adaptability, Self-Learning

EDUCATION

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) Aug 2022 – May 2026

Sushant University, Gurugram, Haryana

Roll No.: 220BTCCSE035

- Specialization in Artificial Intelligence and Machine Learning under the CSE department.
- Coursework includes: Advanced Machine Learning, Deep Learning, Computer Networks, Artificial Intelligence & Knowledge Representation, Data Structures & Algorithms, Database Management Systems, Convex Optimization.
- Final-year major project: Mathematical Steering of LLM Internal States for Safety — a research-grade implementation using RLDA, SVD, Lyapunov stability analysis, and PyTorch hooks for behavioral alignment in transformer models. Near submission for IEEE publication.

EXPERIENCE

Machine Learning Intern — Time Series Forecasting

Jun 2024 – Jul 2024

UpGrad (Remote)

- Designed and developed a robust time-series forecasting model to predict hourly traffic volumes at urban junctions, directly supporting infrastructure planning and urban mobility optimization.
- Engineered complex temporal features from historical datasets and weather patterns, applying lag features, rolling statistics, and Fourier decomposition to significantly improve model accuracy for peak-hour predictions.
- Implemented rigorous cross-validation pipelines and hyperparameter tuning in Python — optimizing learning rate, regularization, and tree depth — to minimize Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) on held-out test sets.
- Conducted extensive Exploratory Data Analysis (EDA) using Pandas and Seaborn to identify traffic anomalies, detect seasonal patterns, and quantify correlation with environmental factors including weather, day-of-week, and holiday effects.
- Delivered a reproducible notebook-based pipeline following MLOps best practices: modular data preprocessing, model versioning, and performance benchmarking across ARIMA, Prophet, and gradient-boosted tree baselines.

RESEARCH & PUBLICATIONS

Mathematical Steering of LLM Internal States for Safety

2026

B.Tech Major Project — IEEE Format (Near Submission)

- Developed a novel framework for mathematically steering the internal representational states of large language models to enforce safe behavioral boundaries, bypassing the need for full RLHF retraining.
- Applied Singular Value Decomposition (SVD) to isolate safety-critical subspaces within transformer attention layers, constructing regularization-based decomposition for alignment (RLDA) objectives.
- Integrated Lyapunov stability analysis to formally verify convergence and behavioral stability of the steering process across adversarial input distributions, providing theoretical safety guarantees.
- Implemented the full experimental pipeline in PyTorch using custom backward hooks to intercept and modify intermediate hidden states at inference time without architectural changes to the base model.
- Conducted ablation studies evaluating steering intensity, subspace rank selection, and trade-offs between helpfulness and safety scores on standard alignment benchmarks.

Unlocking the Potential of Quantum Computers: A Comprehensive Review

2023

Published — IJFMR, Volume 5, Issue 6

DOI: 10.36948/ijfmr.2023.v05i06.9440

- Authored a comprehensive peer-reviewed survey on the theoretical foundations and practical applications of quantum computing, covering quantum gates, superposition, entanglement, and quantum error correction.
- Analyzed the current state of quantum hardware (superconducting qubits, trapped ions, photonic systems) and their viability timelines relative to classical HPC workloads.
- Discussed implications of quantum advantage for AI/ML optimization problems, cryptography, and drug discovery, with structured comparisons of quantum vs. classical algorithm complexity.

Few-Shot Rare Disease Detection — Adversarial Synthesis & Diagnostic Reasoning Framework

2026

Research Implementation Project

- Developed a StyleGAN-based latent-space synthesis pipeline to generate high-fidelity, domain-specific rare disease MRI imagery (Brain Tumors, OASIS dataset) from minimal real samples, addressing critical data scarcity in clinical AI.
- Engineered a Vision Transformer (ViT) anomaly detection model trained on hybrid synthetic-real datasets, achieving superior performance over CNN baselines in low-data regimes through self-attention over spatial patches.
- Integrated a Medical Knowledge Graph module that interprets visual attention maps to recommend confirmatory diagnostic tests (biopsies, genomic sequencing) and structured clinical next steps, creating an explainable AI diagnostic pipeline.

- Technologies: PyTorch, GANs (StyleGAN), Vision Transformers (ViT), Knowledge Graphs, Medical Imaging, Neo4j, OASIS Dataset.

PERSONAL PROJECTS

ATLAS: Advanced Threat & Logistics Analysis System

2025 – 2026

Python, TypeScript, React 18, FastAPI, Neo4j, Pinecone, NVIDIA NIM, Ollama, RAG, Leaflet Full-Stack Risk Intelligence Platform

- Engineered a highly scalable, full-stack risk intelligence platform designed to monitor and map real-time geopolitical supply-chain disruptions across 6 major global maritime chokepoints.
- Architected an asynchronous OSINT data ingestion pipeline utilizing Python, asyncio, and aiohttp to scrape and aggregate unstructured data from over 30 global sources including UN agencies, government bodies, and major news outlets.
- Integrated advanced AI analysis utilizing NVIDIA NIM (GLM-5) and local fallback models via Ollama to evaluate global events, extract structured intelligence, and autonomously score geopolitical threat levels.
- Implemented a sophisticated Retrieval-Augmented Generation (RAG) architecture leveraging Pinecone for cloud-based vector embeddings, in-memory TF-IDF for rapid semantic search, and a local Neo4j knowledge graph to map complex causal relationships and source credibility networks.
- Developed an interactive frontend dashboard utilizing React 18, TypeScript, and Vite, incorporating real-time intelligence feeds, risk trend visualizations via Recharts, and dynamic live world mapping using Leaflet.
- Deployed a robust backend utilizing FastAPI and Uvicorn, integrating SQLite (aiosqlite) for structured relational data storage and SQLAlchemy models with Pydantic for strict data validation and schema enforcement.

NEXUS: Personal Intelligence OS

2025 – 2026

Kotlin, Jetpack Compose, Python, LLaMA.cpp, Whisper.cpp, ML Kit Vision, Room DB Privacy-First On-Device AI Android Application

- Developed a comprehensive, privacy-first 'Personal Intelligence OS' for Android (minSdk 33), serving as a local knowledge base and digital brain with a Material 3 design, single-activity architecture, and a 5-tab layout.
- Engineered a purely on-device AI ecosystem by embedding llama.cpp for local LLM inference (Qwen3.5-2B via GGUF Q4_K_M) and whisper.cpp for speech-to-text processing, ensuring zero cloud dependency and absolute user data privacy.
- Implemented a dynamic chat interface ('Intelligence Hub') utilizing Kotlin and Jetpack Compose, featuring streaming model responses, intent classification, and actionable interaction cards for complex media handling workflows.
- Architected a robust local persistence layer using Room (SQLite) and DataStore to build a high-density 'Knowledge Library' that archives and indexes processed items including YouTube videos, websites, and screenshots.
- Embedded a seamless Python orchestration layer within the Android Kotlin environment utilizing Chaquopy, enabling the execution of complex agent routines, YouTube transcript extraction, and web scraping (beautifulsoup4, requests) directly on the mobile device.
- Integrated ML Kit Text Recognition for on-device OCR, allowing seamless extraction of text from screenshots to feed contextual data into the local LLaMA model for automated summarization and deep work analysis.

Aether: AI-Powered Competitive Intelligence Platform

2025

Next.js 15, React 19, TypeScript, Google Gemini Pro 1.5, Apify, Exa AI

- Built an autonomous, full-stack competitive intelligence application utilizing Next.js 15 (App Router), React 19, and Tailwind CSS v4 to monitor, synthesize, and analyze competitor data across websites and social media platforms.
- Engineered a robust data collection and predictive analytics pipeline leveraging the Apify Client for web scraping, Exa AI for web search, and various APIs (YouTube, Reddit, Twitter) to aggregate unstructured market intelligence data.
- Integrated Google Gemini Pro 1.5 to process collected data, generate strategic recommendations, perform change detection, and evaluate brand sentiment metrics for competitive market positioning.
- Designed a highly performant, serverless backend architecture guaranteeing strict type safety, referential integrity, and rapid querying capabilities at minimal infrastructure cost.

- Implemented a comprehensive authentication system utilizing Better-Auth, supporting Email/Password, Google OAuth, and 2FA/TOTP, alongside automated alert dispatching via Resend (email) and Slack webhooks.
- Enforced strict engineering standards using React Hook Form and Zod for schema validation, Framer Motion for UI transitions, and Shadcn/UI for consistent, accessible component design.

Project Architect Athena: Autonomous Software Engineering Platform

2024 – 2025

Python, Java 11 (JavaFX), FastAPI, Docker, Kubernetes, Multi-Agent Systems, SQLite, JSON Schema, LLMs Polyglot Autonomous Coding Environment

- Architected an experimental, polyglot autonomous software engineering platform featuring a strict separation of concerns between a JavaFX desktop control-plane host and a Python FastAPI orchestration service.
- Developed a deterministic, multi-agent pipeline capable of complex code generation, logical reasoning, and mathematical problem-solving, integrating seamlessly with multiple LLM provider backends.
- Established a robust, schema-driven API boundary utilizing strict JSON Schema contracts to dictate data exchange format and maintain absolute consistency between the Java host and Python backend.
- Engineered a resilient Python backend utilizing FastAPI, Pydantic, and Uvicorn, implementing SQLite-backed run persistence to track agent statuses, execution logs, and generated code artifacts across sessions.
- Built a sandboxed execution environment utilizing Docker and Kubernetes, enabling autonomous agents to write, compile, execute, and validate code safely within an integrated custom IDE setup.
- Automated the entire application lifecycle through unified Bash scripting, streamlining Java Maven dependency management, Python virtual environment bootstrapping, and concurrent service launching.

Text Generation using GPT-2 (Fine-Tuning)

2024

Python, PyTorch, Hugging Face Transformers, NLTK, GPT-2, NLP, Language Modeling, GPU Acceleration

- Engineered a fine-tuning pipeline for the GPT-2 Large Language Model to generate highly coherent, domain-specific text sequences utilizing the WikiText-2 benchmark dataset.
- Architected the model training loop using Hugging Face Transformers and PyTorch, implementing custom DataCollatorForLanguageModeling for dynamic batching and token masking strategies.
- Optimized hyperparameter configurations by defining custom TrainingArguments within the Trainer API, rigorously monitoring model loss and perplexity across sequential training epochs.
- Implemented advanced NLP text preprocessing utilizing NLTK, ensuring pristine data quality by filtering noise prior to tensor vectorization and model ingestion.
- Integrated Kagglehub for seamless, automated dataset acquisition, guaranteeing reproducible training environments across cloud-based GPU instances.

NASA Space Apps Challenge: Exoplanet Prediction — ExoNet

2024

Python, LightGBM, Scikit-Learn, Pandas, SMOTE, K-Fold Cross-Validation, SeabornHackathon Project — 99.21% Predictive Accuracy

- Engineered 'ExoNet,' a high-performance classification pipeline utilizing LightGBM designed to autonomously identify exoplanet candidates from the highly complex Kepler Objects of Interest (KOI) dataset.
- Implemented advanced feature engineering and data cleaning techniques, systematically replacing missing numeric NaNs with medians and programmatically eliminating zero-variance columns via VarianceThreshold.
- Architected robust dataset balancing strategies utilizing SMOTE (Synthetic Minority Over-sampling Technique) to algorithmically synthesize minority class data, massively improving precision on imbalanced exoplanet distributions.
- Developed a rigorous validation framework employing 5-Fold Stratified Cross-Validation to guarantee absolute model stability, completely mitigating statistical flukes or localized overfitting.
- Optimized LightGBM hyperparameter configurations via early stopping callbacks, achieving an exceptional 99.21% predictive accuracy and generating exhaustive confusion matrices utilizing Seaborn and Matplotlib.

Medical Image Classification

2024

Python, TensorFlow, Keras, OpenCV, Pillow (PIL), CNNs, Deep Learning, TensorBoardEnd-to-End Clinical Vision Pipeline

- Architected an end-to-end deep learning vision pipeline employing custom Convolutional Neural Networks (CNNs) to accurately diagnose and classify complex patterns in high-resolution medical imagery.

- Developed a robust model training architecture utilizing TensorFlow and Keras, employing sequential dense layers, pooling, and dropout mechanics to prevent catastrophic overfitting on specialized clinical datasets.
- Engineered scalable data ingestion algorithms utilizing Keras Image Data Generators to efficiently manage memory overhead and execute real-time batch processing during neural network training.
- Integrated TensorBoard monitoring and advanced Keras Callbacks (EarlyStopping, ModelCheckpoint) to actively track validation loss, epoch accuracy, and automatically persist optimal .h5 model weights.
- Built dedicated inference validation scripts utilizing OpenCV and Pillow (PIL) to algorithmically preprocess raw, unseen imagery to match strict neural network input tensor requirements.

Airline Tweet Sentiment Analysis

2024

Python, NLTK, spaCy, Scikit-Learn, TextBlob, Pandas, TF-IDF, Logistic Regression, NLP NLP Classification Pipeline

- Engineered a robust Natural Language Processing (NLP) pipeline to autonomously extract, process, and classify sentiments from unstructured airline customer review tweets at scale.
- Developed extensive text preprocessing routines utilizing NLTK and spaCy to execute advanced tokenization, stop-word removal, and WordNet lemmatization for clean feature representation.
- Implemented TF-IDF (Term Frequency-Inverse Document Frequency) vectorization using Scikit-Learn to mathematically transform raw text strings into highly predictive numerical feature matrices.
- Built and trained a Logistic Regression classification model to accurately categorize positive, neutral, and negative sentiment polarities with aggressive hyperparameter tuning to reduce classification variance.
- Conducted comprehensive Exploratory Data Analysis (EDA) using TextBlob and custom Regex patterns to systematically resolve noise, emojis, and irregular social media syntax patterns.

IT Support Ticket Classifier

2024

Python, Scikit-Learn, TF-IDF, Logistic Regression, NLP, Machine Learning Pipelines Enterprise NLP Automation Tool

- Developed an automated NLP classifier designed to autonomously interpret and route enterprise IT support tickets (password resets, leave balances, hardware requests) without human triage.
- Engineered a streamlined machine learning Pipeline utilizing Scikit-Learn, combining rigorous TfidfVectorizer text transformations with a scalable LogisticRegression predictive model.
- Architected a highly efficient mathematical vectorization strategy to parse short-form user queries, significantly minimizing manual triaging and drastically improving organizational response times.
- Implemented modular classification logic that evaluates token combinations and semantic context, enabling the system to predict correct support categories with high precision over localized enterprise datasets.

CNN MNIST Digit Recognition

2023

Python, TensorFlow, Keras, CNNs, Matplotlib, NumPy, Deep Learning Foundational Deep Learning Implementation

- Built a highly accurate Convolutional Neural Network (CNN) utilizing TensorFlow and Keras to autonomously classify handwritten numerical digits from the industry-standard MNIST dataset.
- Engineered mathematically rigorous data preprocessing sequences involving multi-dimensional tensor reshaping and strict pixel intensity normalization (0–255 scale) to accelerate gradient descent convergence.
- Implemented complex one-hot encoding methodologies utilizing to_categorical to mathematically structure classification targets for efficient loss calculation via categorical cross-entropy.
- Architected a multi-layer sequential neural network integrating 2D Convolutional layers, Max Pooling, and Dense output layers with Softmax activation to compute highly calibrated class probabilities.
- Evaluated deep learning model performance via dedicated training and validation splits, generating comprehensive loss graphs and accuracy curves utilizing Matplotlib.

Diabetes Prediction System

2023

Python, Scikit-Learn, Pandas, SVM, Random Forest, Logistic Regression, ROC-AUC Healthcare Predictive Analytics

- Developed a comprehensive, end-to-end predictive healthcare system utilizing supervised machine learning to accurately predict diabetes onset based on complex physiological feature datasets.
- Engineered rigorous EDA and cleaning pipelines utilizing Pandas, programmatically addressing missing values, handling duplicate records, and structuring numeric matrices for algorithmic ingestion.

- Implemented strict mathematical feature scaling using StandardScaler to normalize standard deviations and means, drastically improving convergence stability across distance-based classification algorithms.
- Architected and cross-evaluated multiple classification models including Support Vector Machines (SVC), Random Forests, and Logistic Regression to empirically determine the optimal algorithm for the dataset.
- Validated model efficacy utilizing Scikit-Learn's cross_val_score, calculating Receiver Operating Characteristic (ROC) curves, and deriving comprehensive Area Under the Curve (AUC) metrics for clinical reliability.

ML Algorithms from Scratch (K-Means, KNN, SVM)

2023

Python, NumPy, Matplotlib — No ML Library Dependencies

Low-Level Algorithm Engineering

- K-Means Clustering: Engineered a pure mathematical implementation of K-Means Clustering entirely from scratch utilizing NumPy, implementing centroid initialization, Euclidean distance computation, and iterative convergence loops without any ML library abstractions.
- K-Nearest Neighbors (KNN): Built a robust, pure NumPy implementation of KNN classification, engineering efficient distance-calculation functions and majority-voting logic to assign class labels from K closest training vectors.
- Support Vector Machine (SVM): Architected an advanced SVM binary classifier from scratch utilizing raw NumPy — engineering a stochastic gradient descent (SGD) training loop with manual derivative calculations, hinge loss regularization, and hyperplane margin maximization.
- Visualized iterative algorithm transitions, decision boundaries, and support vectors utilizing Matplotlib to formally prove algorithmic correctness and convergence behavior.

Titanic Survivor Predictor

2023

Python, Scikit-Learn, Pandas, NumPy, Decision Trees, Label Encoding

Introductory Predictive Analytics

- Built an introductory predictive analytics model employing Decision Trees to mathematically estimate passenger survivability likelihoods based on the historical Titanic dataset.
- Engineered extensive categorical feature transformations, executing LabelEncoder routines to convert string-based categories (Sex, Embarked) into algorithmically ingestible numerical integers.
- Developed structured data splits utilizing Scikit-Learn's train_test_split to rigorously separate testing and training nodes, ensuring the Decision Tree maintained generalized predictive integrity.

ACCOMPLISHMENTS & CERTIFICATIONS

Hackathon & Competition Awards

- 1st Prize — Paper Presentation, International Technical Symposium AURA 25 (International Level).
- Mumbai Hacks HealthTech Finalist — Ranked 3rd nationwide in the HealthTech Track, selected for the prestigious final in-person round in Mumbai.
- 3rd Place — Masters Union GHOST OPS: FOCOS Strategy Hackathon (Team SYNAPSE).
- NASA Space Apps Challenge Participant — Developed ExoNet achieving 99.21% exoplanet classification accuracy.

Academic & Leadership Achievements

- 2nd Place — Inter-School MUN, Gurugram (2017).
- 1st Place — Inter-House DAV MUN (2018).
- 2nd Place — Inter-House DAV MUN (2019).
- 1st Place — Inter-House Football Championship (2015).
- 2nd Place — Inter-House Football Championship (2017).
- 2nd Place — Inter-School Football Championship (2018).

Certifications

- Data Science Master Course — Coding Blocks (Certificate No.: CBOL-295863-7618f3f4).
- Full Stack Web Development — Coding Blocks (Certificate No.: CBOL-299976-a5884660).
- Workshop on Tableau & Data Visualization — UpGrad Campus.

Publications

- 'Unlocking the Potential of Quantum Computers: A Comprehensive Review' — Published in IJFMR, Volume 5, Issue 6, 2023. DOI: 10.36948/ijfmr.2023.v05i06.9440.

ABOUT & ADDITIONAL CONTEXT

Preekshit Saklani (online alias: cryptic-ai) is a 21-year-old AI/ML engineer, researcher, and builder based in Gurugram, Haryana, India, currently completing his final year of B.Tech in CSE (AI) at Sushant University. His engineering philosophy centers on building production-grade, resource-efficient AI systems — a conviction rooted in working with constrained hardware (NVIDIA RTX 3050, 4GB VRAM, 16GB RAM, Fedora Linux) and a commitment to making intelligence accessible without expensive cloud infrastructure.

He is also in the ideation and early execution phase of TensorForge Systems — a B2B AI data infrastructure startup designed to deliver autonomous data ingestion, cleaning, and delivery pipelines (DPaaS), with WORM-compliant immutable storage for institutional clients. Outside of engineering, Preekshit is passionate about football, Formula 1, chess, Valorant, Indian military history, personal finance, and long-term investing.

He consistently gravitates towards zero-cost or free-tier toolchains, on-device model inference, multi-database architectures (combining SQL, NoSQL, vector, and graph databases within a single system), and autonomous multi-agent frameworks. His preferred aesthetic in engineering artifacts is institutional, formal, and powerful — reflecting the same rigor he applies to his technical work.